

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1288

Howard Beach 100 kV
October 26, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1288. The Interconnection Customer proposes a 53.3 MW solar photovoltaic facility interconnecting at the Howard Beach 100 kV substation in the Ohio area. The request was received on September 12, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 471 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$50,041,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Tidd – Muskingum Riv 500.0 kV L, the Brooks Hill – Oreg 3 Inc 230.0 kV loads to 115.9% of its 405 MVA rating. Project DFAX: 15.3%.

Myrtle-Date – Brooks Hill 230.0 kV: 112.4% loading (437 MVA) following the outage of the Tidd – Muskingum Riv 500.0 kV; DFAX = 15.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Brooks Hill – Oreg 3 Inc 230.0 kV	Rebuild line with higher-capacity conductor	\$8,743,000	100% (\$8,743,000)
Myrtle-Date – Brooks Hill 230.0 kV	Rebuild line with higher-capacity conductor	\$41,298,000	100% (\$41,298,000)
	Total allocated to project		\$50,041,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.